Radical-pair compass: a simulation-only navigation-relevance feasibility study

Question. Under what conditions, if any, could a radical-pair (spin-chemical) magnetic compass be useful as an auxiliary heading sensor for GNSS-denied inertial navigation, compared with a conventional MEMS magnetometer? A "no, never" answer was declared a valid outcome in advance.

Scope. Everything here is simulation. No experiment exists and no new physics is claimed. Parameters marked ASSUMED are neither measured nor verified; they are swept. The attitude (not position) filter is the navigation object studied. Pre-registration: `PLAN.md` (committed before any run) plus dated amendments; every deviation is in `CHANGELOG.md`.

What is and is not new

Not new: radical-pair compass precision limits and Fisher-information analyses (Ren, Hiscock & Hore 2021; Procopio & Ritz 2013, 2020; Cai 2011; Bezchastnov & Domratcheva 2026), cryptochrome magnetic sensitivity (Xu et al. 2021), the ChemNav programme, and flight-tested quantum magnetic navigation (arXiv 2504.08167, anomaly matching, not compassing). A web search found no study that models a radical-pair sensor as an aiding measurement in an inertial filter; that is absence of evidence from a limited search (one patent, US12078486, was read: low overlap). **New (the only claim):** an open, reproducible requirements-and-feasibility framework that connects a spin-yield model to a photon-limited estimator and an attitude-filter comparison against a MEMS magnetometer. Unverified citations are listed in `open_questions.md`.

Summary of findings

1. **Spin model.** With verified literature tensors (Hiscock 2016), the minimal two-nitrogen model has a yield anisotropy of 0.206; assumed richer models give about 0.01 and relaxation at 10^7/s leaves 1.5x the 10^{-3} threshold. The minimal model is an optimistic upper bound.

2. **Sensor.** The yield is symmetric under field reversal, so only the field *axis* is observable. Because both nitrogen tensors share one principal frame the molecule has exact sign-flip symmetries: element sets related by axis permutations have a four-fold mirror ambiguity (failure rate 0.71, pre-registered Gate S2 FAILED). A non-orthogonal set chosen by a pre-registered rule removes it (Gate S2' passed on fresh data).

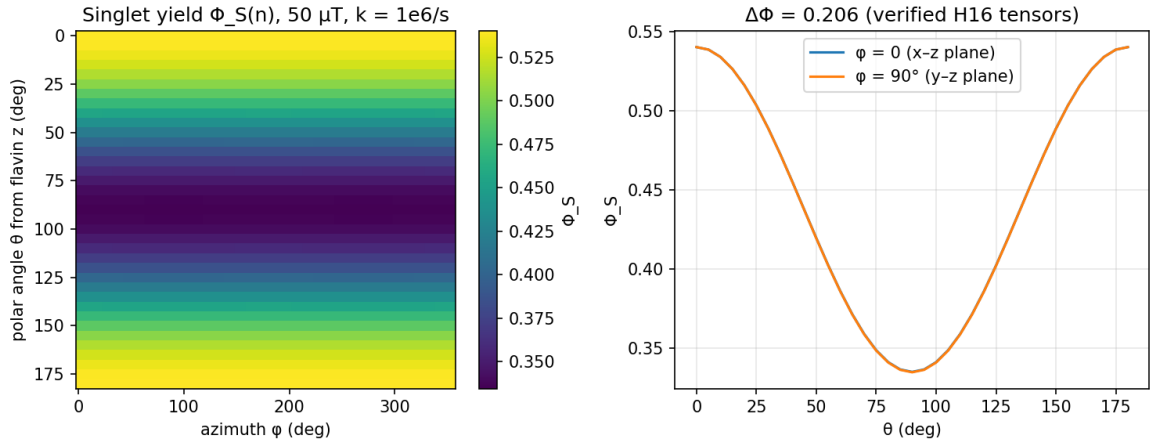
3. **Navigation, pre-registered:** 0 of 390 configurations reduce yaw RMSE by at least 20% at both 60 s and 600 s relative to a MEMS magnetometer: **answer NO**. The baseline filter was badly inconsistent (unmodelled soft iron), and Gate S3(i) FAILED.

4. **Navigation, with online calibration (Amendment 3, specified after the NO):** filters become consistent; 12 of 270 configurations meet the criterion, mostly in two corners (a noisy 2 uT magnetometer against a 0.3 deg/axis radical-pair sensor, and 15 deg dip; one D row sits at a high-bias, low-ARW gyro). At 0.5 uT and 45 deg dip none do.

5. **Photons are not the binding constraint** for a 0.3 deg/axis sensor (see requirements); the shared magnetic disturbance, the baseline's calibration and the sensor-error model are.

1. Spin-dynamics model (Stage 1)

Radical A carries two spin-1 nuclei (N5, N10), radical B none; B_0 = 50 uT, singlet-born, Haberkorn recombination $k_S = k_T = k = 10^6$/s. Yield: $\Phi_S(n) = (1/M)\ \Sigma\ |\langle m|P_S|n\rangle|^2\ k^2/(k^2 + (E_m - E_n)^2)$. **Tensors:** the values recalled in the study brief did not match the primary source; the verified values (N5 = (-0.087, -0.100, 1.757) mT, N10 = (-0.014, -0.024, 0.605) mT, Hiscock et al. PNAS 2016) are primary, the recalled set is a labelled sensitivity set. **Validation (Gate S1 PASSED):** isotropic hyperfine gives a direction-independent yield (spread 1e-15); H = 0 gives yield 1; yields in [0,1]; $\Phi(n) = \Phi(-n)$; an independent Liouville-space implementation agrees to 1e-15 on 200 directions; grid and surrogate converge. Primary anisotropy $\Delta\Phi$ = 0.2057 (Φ from 0.334 to 0.540), against the pre-registered threshold 10^{-3}.



Model (50 uT, k = 1e6/s)	ΔΦ	margin vs 1e-3
minimal H16 (primary)	0.2057	205.7x
R1 a_iso_B=0.0 d_B=0.1	0.0977	97.7x
R1 a_iso_B=1.5 d_B=0.5	0.0131	13.1x

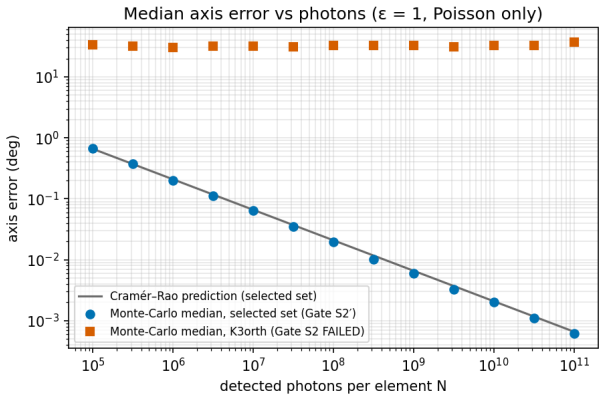
Model (50 uT, k = 1e6/s)	ΔΦ	margin vs 1e-3
R2	0.0099	9.9x
R3 (EXPLORATORY)	0.0109	10.9x
(exploratory) exchange J_mT=0.5	0.0013	1.3x
(exploratory) dipolar D_mT=0.3	0.0108	10.8x
(exploratory) relaxation relax_rate=100000.0	0.1707	170.7x
(exploratory) relaxation relax_rate=1000000.0	0.0626	62.6x
(exploratory) relaxation relax_rate=10000000.0	0.0015	1.5x

R1-R3 and the exchange, dipolar and relaxation terms are ASSUMED parameters (no verified Trp-side tensors were available); relaxation values come from a meridian scan.

2. Photon-limited sensor and estimator (Stage 2)

K identical oriented elements at rotations R_j; counts ~ Poisson(N(1 + ε(Φ_j - Φ■)/Φ■)); ε = 1 is an ideal ordered ensemble, ε < 1 absorbs partial order and incomplete yield transfer (ASSUMED). Estimator: Poisson maximum likelihood over the field axis (hemisphere grid plus multi-start Fisher scoring); bound: analytic Cramer-Rao in the tangent plane.

Gate S2 (pre-registered, K3orth) FAILED: ratios of Monte-Carlo to bound 14,478, 25,621, 51,653, slope -0.013. Cause: exact sign-flip symmetry (the likelihood has equal-height mirror modes), so a local bound is ill-posed. **Amendment 1** (post hoc, committed before any new run) selected a K=3 set by maximising the minimum Poisson KL divergence between axes more than 10 deg apart (design seeds only), then evaluated it on fresh data: **Gate S2' PASSED** (ratios 0.971, 0.935, 0.937, slope -0.523). Amendment 1.1 fixed a blind spot of the first objective (found by a smoke test before any design run); a single-start estimator bug was found after S2' passed and fixed, with all results re-run (CHANGELOG).

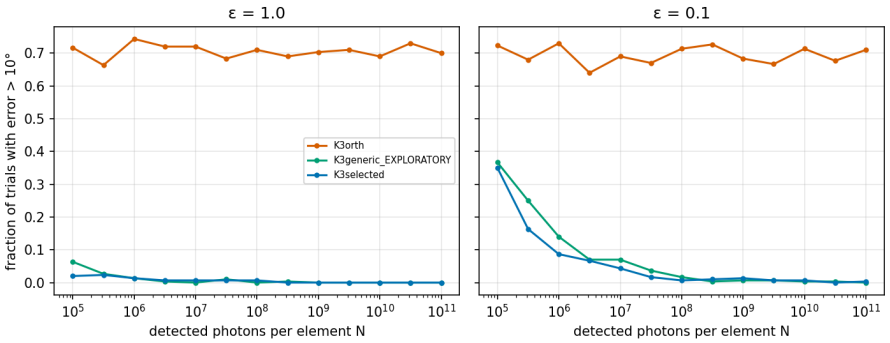


Does the ambiguity need a shared principal frame? Yes. Tilting the N10 frame by 5 deg makes mirror images separable (D = 10 nats at the weakest pair) at 4.6x10^6 photons in the minimal model; in a richer assumed model with a 15 deg tilt the ambiguity survives at 10^8 photons (needs 1.7x10^8). Tilts are ASSUMED.

Contrast row	ε	N for 5 deg (median dir)	N for 5 deg (worst dir)	N for failures ≤ 5%	N for 0.3 deg/axis (A2.2)
ideal	1	1.5x10^3	6.1x10^7	3.2x10^4	3.1x10^5
0.1	0.1	1.6x10^5	7.2x10^9	3.2x10^6	3.2x10^7
richer model (derived)	0.0435	8.4x10^5	3.8x10^{10}	3.2x10^7	1.7x10^8
relaxation 1e6 (derived)	0.388	1.0x10^4	4.5x10^8	3.2x10^5	2.1x10^6
relaxation 1e7 (derived)	0.0111	1.3x10^7	5.9x10^{11}	3.2x10^8	2.6x10^9

Photons per element per measurement, selected K=3 set. Median-direction values use the Cramer-Rao median error; the failure column is the smallest N with at most 5% of trials more than 10 deg off (300 trials, half-decade grid); worst-direction values show that near-singular field directions need orders of magnitude more photons. Derived contrasts are ratios of relative anisotropies from Stage 1 (approximate). At the Cramer-Rao N for a 5 deg median error, 24-26% of Monte-Carlo trials still fail by more than 10 deg (results/stage2_requirements_mc.csv): the bound alone is optimistic at low contrast.

Comparison with Ren et al. (2021): with their phenomenological signal and K=1000 random orientations our bound gives 6.8x10^7 photons (default) against their 5.3x10^7-2.3x10^8, and 4.6x10^6 (optimised) against 3.94x10^5: a factor of about 12 remains and is not tuned away. With only K=3 elements the gap was ~800x; element count explained most of it.


ε = 1.0
ε = 0.1
fraction of trials with error > 10°
detected photons per element N
detected photons per element N
K3orth
K3generic_EXPLORATORY
K3selected

3. Attitude-filter experiment (Stage 3)

Multiplicative EKF (quaternion, nav-frame error state, 6 states: attitude error and gyro bias) at 10 Hz on a 100 Hz truth; 600 s scripted trajectory (3-5 deg tilt oscillation, three slow +/-90 deg yaw turns, then a figure-eight, tilt up to 7 deg); gyro bias instability {1, 10, 100} deg/h, ARW {0.1, 0.3, 1} deg/sqrt(h); accelerometer 0.05 m/s^2 plus 0.3 m/s^2 white vibration; Earth field 50 uT at dip {15, 45, 66} deg. **Arms:** A gyro + accelerometer; B + MEMS magnetometer (full vector, noise {0.1, 0.5, 2} uT); C + radical-pair axis (random sign, isotropic tangent error sigma_rp {0.3, 1, 2, 5, 10} deg, sign-invariant residual); D = B + C. **Disturbance, identical for both sensors** (no immunity is credited): 1% soft iron, hard iron and motor-current-correlated fields with nominal RMS sigma_d {0, 2, 10} uT. 200 paired runs per configuration (common random numbers). Decision criterion (pre-registered): yaw-RMSE reduction of at least 20% at both 60 s and 600 s with a paired-bootstrap 95% lower bound above 0.

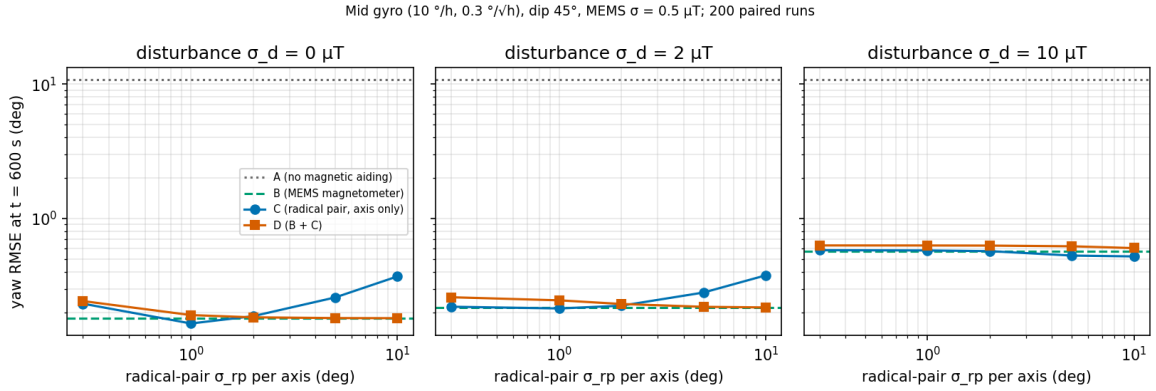
Validation gates

Gate	Status	Detail
S1	**PASSED**	checks a-f pass; primary anisotropy 0.2057 vs threshold 0.001
S2 (pre-registered)	**FAILED**	condition 1 ratios 1.448e+04, 2.562e+04, 5.165e+04 (need 0.5-2); condition 2 slope -0.013 (need -0.5 +/- 0.1)
S2' (amended, Amendment 1)	**PASSED**	condition 1 ratios 0.971, 0.935, 0.937 (need 0.5-2); condition 2 slope -0.523 (need -0.5 +/- 0.1); condition 3: Ren comparison documented in results/stage2_sprime.json
S2' diagnostic (not a gate)	**DIAGNOSTIC**	ratios 0.928, 0.892, 0.872; slope -0.488
S3(i) NEES (pre-registered)	**FAILED**	outside the family-wise 95% interval [5.32, 6.73]: arm A at 600 s: 7.24, arm C at 600 s: 6.83; 10 of 12 tests inside
S3(i) diagnostic (not a gate)	**DIAGNOSTIC**	12 of 12 tests inside the same interval
S3(i)' (amended, POST HOC)	**PASSED**	12 of 12 tests inside [5.32, 6.73]; not an independent test
S3(ii) arm A vs analytic drift	**PASSED**	ratio simulated/analytic 1.028 to 1.063 (need within a factor 2)
S3(iii) independent filter	**PASSED**	max |yaw difference| 1.19e-08 rad (need < 1e-6)
S3 overall (pre-registered)	**FAILED**	(i) failed
S3' overall (amended, post hoc)	**PASSED**	reported next to, never instead of, S3

Gate S2 (K3orth) and S3 stay FAILED. S2' and S3(i)' were defined after the failures with the owner's approval; S3(i)' is post hoc and its outcome was visible beforehand, so it is a confirmation of the diagnosis (unmodelled figure-eight acceleration absorbed by the bias states), not an independent test.

4. Results

Pre-registered answer: NO. 0 of 390 C/D configurations meet the criterion. At 60 s no configuration exceeds 3.5%; at 600 s alone 20 pass, all of them artefacts: in the main grid the MEMS filter is strongly over-confident (NEES 183-480 against about 6 even with no disturbance) because soft iron is not in its noise model, so a sensor that trusts the same biased measurement less appears to win. Any magnetic aiding helps hugely (yaw RMSE at 600 s: 10.8 deg gravity-only against 0.18-0.57 deg), and the radical-pair channel adds almost nothing on top of it under shared disturbance.



Exploratory E1/E2. Adding the soft-iron variance to R did not restore consistency (NEES still 118-462) because the error is a constant body-frame bias: criterion met in 0 of 90. With no soft iron and no disturbance the filter is consistent, and a sensor twice as accurate as the MEMS one (0.3 deg against about 0.57 deg) improves yaw by only 3% (60 s) and 11% (600 s): criterion met in 0 of 30.

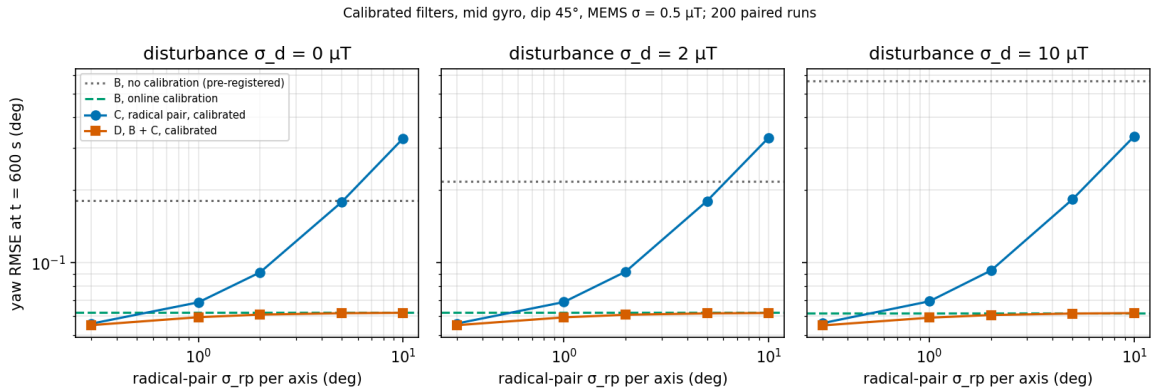
Online calibration (Amendment 3)

Both sensors get the same treatment: 18-state filters that estimate hard iron (3), soft iron (6, symmetric) and the motor-current field per unit throttle (3, throttle profile known), with priors consistent with the truth generator and no inflation of R. Filters become consistent (NEES about 6-7 even at sigma_d = 10 uT) and the magnetometer baseline improves (e.g. 600 s yaw RMSE 0.22 deg to 0.06 deg). **12 of 270 configurations now meet the criterion**:

Arm	MEMS noise (uT)	dip	sigma_d	sigma_rp	60 s: reduction (CI low)	600 s: reduction (CI low)
D	2.0	45	0.0	0.3	28% (20%)	47% (40%)
C	2.0	45	2.0	0.3	26% (15%)	46% (40%)
D	2.0	45	2.0	0.3	46% (39%)	46% (40%)
C	2.0	45	10.0	0.3	49% (41%)	46% (40%)
D	2.0	45	10.0	0.3	60% (55%)	46% (40%)

Arm	MEMS noise (uT)	dip	sigma_d	sigma_rp	60 s: reduction (CI low)	600 s: reduction (CI low)
D	2.0	45	10.0	1.0	31% (24%)	36% (29%)
C	0.5	15	0.0	0.3	45% (37%)	21% (12%)
D	0.5	15	0.0	0.3	38% (33%)	25% (18%)
C	0.5	15	2.0	0.3	50% (41%)	21% (11%)
D	0.5	15	2.0	0.3	48% (43%)	25% (18%)
D	0.5	15	10.0	0.3	56% (53%)	24% (17%)
D	0.5	45	2.0	0.3	24% (18%)	26% (21%)

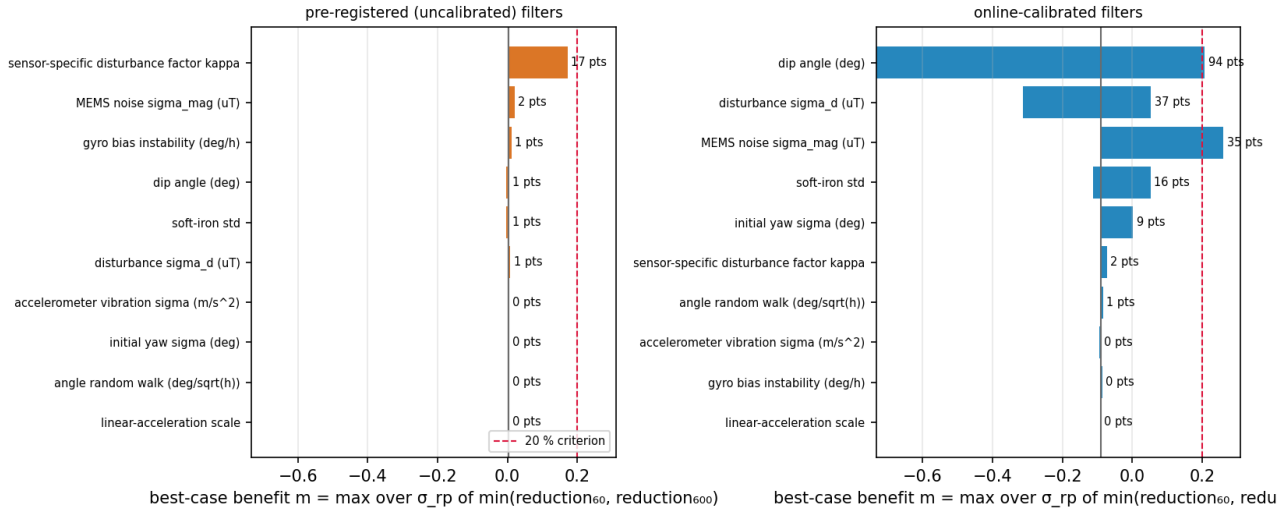
Mid gyro (10 deg/h, 0.3 deg/sqrt(h)) unless gyro corners are shown in the CSV. All passing rows have sigma_rp = 0.3 deg (the smallest grid value) except one D row at 1 deg, so the benefit at smaller sigma_rp is not known. At the base setting (0.5 uT, 45 deg) arm C (0.3 deg) changes yaw RMSE by -9% (60 s) and +10% (600 s) at sigma_d = 2 uT (-31% and +10% at sigma_d = 0) relative to the calibrated magnetometer: calibration of an axis-only measurement converges more slowly. The 12 passes cluster in structured regions with lower bounds of at least 11%, so they are unlikely to be multiplicity noise, but 270 configurations were tested.



Observability. Gravity-only has numerical rank 5 (the single null direction is yaw); arms B, C and D have rank 6 in hover and turn windows at dips 15, 45 and 66 deg, so the axis-only channel loses nothing locally (information-implied yaw std at 45 deg hover: B 0.113, C 0.131, D 0.110 deg). **Global sign ambiguity** (exploratory snapshot initialisation): the axis-only sensor flips yaw by 180 deg in at most 6.7% of draws (dip 15, sigma_rp 10 deg, no disturbance) and essentially never at 45 deg dip. **Sensor-specific advantage kappa** (exploratory, corrected model): the radical-pair channel helps only if it were assumed to respond to the disturbance much less than a MEMS magnetometer; nothing in this study supports that. (kappa = 0.5, sigma_d = 10 uT, arm C: criterion met in 2 of 2.)

5. Sensitivity (Stage 4)

Every ASSUMED parameter was varied one at a time around the base (mid gyro, dip 45 deg, 0.5 uT, sigma_d = 2 uT) for both filter types; the metric is the best-case benefit m = max over sigma_rp of min(reduction at 60 s, reduction at 600 s) of arm C relative to B. Base m: pre-registered filters 0%, calibrated -9%.



Rank	Parameter (calibrated)	swing of m (points)	reaches 20%?	Parameter (pre-registered)	swing (points)	reaches 20%?
1	dip angle (deg)	94	yes	sensor-specific disturbance factor kappa	17	no
2	disturbance sigma_d (uT)	37	no	MEMS noise sigma_mag (uT)	2	no
3	MEMS noise sigma_mag (uT)	35	yes	gyro bias instability (deg/h)	1	no
4	soft-iron std	16	no	dip angle (deg)	1	no
5	initial yaw sigma (deg)	9	no	soft-iron std	1	no
6	sensor-specific disturbance factor kappa	2	no	disturbance sigma_d (uT)	1	no
7	angle random walk (deg/sqrt(h))	1	no	accelerometer vibration sigma (m/s^2)	0	no

Rank	Parameter (calibrated)	swing of m (points)	reaches 20%?	Parameter (pre-registered)	swing (points)	reaches 20%?
8	accelerometer vibration sigma (m/s^2)	0	no	initial yaw sigma (deg)	0	no

6. Limits, what was not modelled, and what was assumed

Not modelled: position and velocity (no GNSS-denied position drift); real magnetic environments (gradients, anomalies, motor EMI spectrum, time-varying external fields); chemistry kinetics, temperature (the closest experiment, Kerpal 2019, was at 120 K), light level, photobleaching, ageing; orientational disorder inside the spin model (only through epsilon); hyperfine fluctuations; direction-dependent radical-pair covariance (isotropic Gaussian used) and ambiguity outliers (up to 5% not simulated); correlation of the two sensors' errors in the uncalibrated arm D; accelerometer bias and scale errors, gyro scale factor and misalignment, coning, timing and latency; hardware of any kind.

ASSUMED parameters (all swept or fixed as stated in PLAN.md): hyperfine principal-axis alignment and all Trp-side couplings; exchange, dipolar and relaxation values; contrast epsilon and its derived values; lock-in efficiency 2/pi and 10 Hz updates; background, electronics noise and gain drift; photon caps; gyro bias/ARW and the 10% bias random walk; accelerometer noise and vibration; initial errors (1, 1, 10 deg); Earth field 50 uT and dip; MEMS noise; soft-iron 1%; disturbance split, throttle profile and nominal RMS; radical-pair per-axis error sigma_rp; kappa = 1; calibration priors; trajectory.

Weakest points: (1) no measurement anywhere: epsilon and sigma_rp are assumptions; (2) the calibrated baseline is an idealised EKF with exact priors and known throttle profile, which flatters both sensors; (3) a sequence of post-hoc amendments after failed gates, even though each is dated, committed first and keeps the failure on record. Reproduction: `README.md`; seeds from SeedSequence(20261013); Python 3.14, numpy 2.4, scipy 1.17.